IRB Study Number:
Version #, Date:

1) Protocol Title

Validation of a ankle diagnostic devic

2) IRB Review History*

N/A.

3) Objectives*

To test the accuracy of a newly developed ankle diagnostic machine.

*Background**

Ankle diagnostics have been typically performed using subjective analysis. This study will examine the validity and feasibility of use of a newly developed ankle diagnostic machine that can quantify limitations throughout an ankle range of motion.

4) Inclusion and Exclusion Criteria*

To be included in the study subjects:

- Must be a man or woman between 18 and 60 years of age;
- has a previous self-reported injury of a lateral or medial ankle sprain confirmed by an athletic trainer or physical therapists examination
- may not have any uncontrolled neuromuscular or cardiovascular disease;
- must be capable of ambulation for at least 50 feet with or without an assistive device and able to get up and down from the floor with minimal assistance;
- May not have any metal inclusions or prior surgical history for the tested ankle.

None of the following special populations will be included in the study:

- Adults unable to consent
- Individuals who are not yet adults (infants, children, teenagers)
- Pregnant women
- Prisoners

5) Number of Subjects*

100 subjects will be recruited, and all assessments will be performed at the Max Orovitz Laboratory Complex on the Coral Gables campus.

6) Study-Wide Recruitment Methods*

Subjects will be recruited through flyers and personal communications.

7) Study Timelines*

Subjects will attend a single session. First the subject's ankle will be correctly aligned in the machine. Next the subject will perform three practice trial during which they will attempt to trace a predetermined circular path with the laser included on the anterior portion of the machine's footplate. Following these practice trial the subject will perform three test trial during which data will be collected using the machine's self-contained variable resistors to determine foot position.

8) Study Endpoints*

Patients will attend a single session.

9) Procedures Involved*

Upon recruitment the subject will come to the Max Orovitz laboratory complex at 1507 Levante Ave. They will be asked to complete a questionnaire detailing their ankle injury and perceived limitations. The subject's reported limitations will then be assessed by a physical therapist or athletic trainer using the Talar Tilt, Talar Tilt-D, Anterior Draw and External Rotation Stress tests detailed in the Table below. The subject's ankle will then be correctly aligned in the machine with the knee at $110^{o}\pm10^{o}$.

Next the subject will perform three practice trials during which they will attempt to trace a predetermined circular path with the laser included on the anterior portion of the machine's footplate. Following these practice trial the subject will perform three test trial during which data will be collected using the machine's self-contained variable resistors to determine foot position.

Pictures and videotaping may occur during testing if the subject completes the HSRO Audio/Video/Photography Authorization form.

The following are to be completed by the Athletic Trainer/Physical Therapist:

Special Tests	**Positive Test (+)**		**Negative Test (-)**		**Comments**
	L	R	L	R	
Talar Tilt (Calcaneofibular Ligament) *Patient Position*: supine *Instructions*: The examiner stabilizes the patient's ankle in neutral and stresses it into inversion. A positive test is laxity when compared bilaterally					
Talar Tilt-D (Deltoid Ligament) *Patient Position*: supine *Instructions*: The examiner stabilizes the patient's ankle in neutral and stresses it into eversion. A positive test is laxity when compared bilaterally					

Anterior Drawer (Anterior Talofibular Ligament) *Patient Position*: supine *Instructions*: The examiner stabilizes the foot in 20 degrees of plantarflexion and translates the talus anteriorly on the tibia. A positive test is laxity when compared bilaterally					
External Rotation Stress Test (Syndesmotic Injury) *Patient Position*: seated at edge of table with knee flexed to 90 degrees *Instructions*: The examiner stabilizes the leg with one hand, while holding the ankle at 90 degrees and simultaneously places the ankle into external rotation. A positive test is pain.					

10) Data and Specimen Banking*

Data or specimens banking are not part of the study.

11) Withdrawal of Subjects*

Subjects may be removed from the study due to illness or injury or for failure to attend two scheduled appointments without providing 24-hour notice. They may also withdraw voluntarily at any point during the study timeline.

12) Risks to Subjects*

The potential risks that might expected from the testing include:

- Mild discomfort at the ankle when the subject is performing the circular movement.
    - Subjects will be told they have the right to ask any questions about possible hazards involved with this study at any time.

12) Potential Benefits to Subjects*

Subjects will derive no benefits from the test; however the test may become a diagnostic tool for quantifying ankle pathologies.

13) Vulnerable Populations*

All subjects will be a minimum of 18 years of age. No pregnant women, prisoners, children or cognitively impaired individuals fit the inclusion criteria.

Students: If the potential subject is a student, he or she will be advised that the decision to participate or not to participate, or to withdraw from the study, will not affect his or her grades or other academic standings within the University.

Employees: If the potential subject is an employee of the University, he or she will be advised that the decision to participate or not to participate, or to withdraw from the study will not affect their employment within the University.

14) Multi-Site Research*

All testing sessions will be conducted at the Neuromuscular Research and Active Aging Laboratory located at the Max Orovitz building on the University of Miami Coral Gable campus.

15) Community-Based Participatory Research*

N/A

16) Sharing of Results with Subjects*

Subjects will be informed of the results of the study.

17) Setting

Training - The location of the training and research is University of Miami, Max Orovitz laboratory, 1507 Levante Ave, Coral Gables.

18) Resources Available

Dr. Signorile has been conducting research on exercise with older persons for the past 25 years. Dr. Aldousany and Dr. Reddy are a licensed Physical Therapist and certified athletic trainer. All research assistance will be third and fourth year and graduate students in Kinesiology.

19) Prior Approvals

N/A

20) Recruitment Methods

Subjects will be recruited using flyers and personal communication. Flyers will be placed around the campuses of the University of Miami. There will be no remuneration provided for the subjects. Screening to ascertain eligibility will be done by licensed Physical Therapists.

21) Local Number of Subjects

100 men and women between the ages of 18 and 90 will be incorporated in the study.

22) Provisions to Protect the Privacy Interests of Subjects

None of the forms will have their personal information, and will be identified solely by subject number. Also, the information requested will be clearly conveyed in the informed consent forms. Subjects will also be made aware that they can deny any tests or information and their data will be secured. Subject data will be stored in a locked cabinet in the Laboratory of Neuromuscular Research and Active Aging and only the Principal Investigator and research staff will have access. Any electronic data will be stored in password locked files with access only by the Principal Investigator and research staff. As a further safeguard all data sheets and files will have only subject numbers as identifiers and the key to break that coding will be stored in a separate locked file cabinet in the main office of the laboratory with only the Principal Investigator and the laboratory administrator having access. All photographs or filmstrips collected on subjects that have completed the HSRO Audio/Video/Photography Authorization form will be stored in the same password locked files as their data.

23) Compensation for Research-Related Injury

If injury occurs during the study medical assistance will be available. In these cases, either the subject or his or her insurance carrier will be responsible for all costs.

24) Economic Burden to Subjects

There will be no costs associated with participation in this study and all training will be conducted at no charge to the participant.

25) Consent Process

The PI and study coordinator will attain the informed consent for individual subjects. This will be completed before any testing or other procedures are initiated. The form will be written and explained in the English language on the first day of introduction within the confines of the laboratory. Only English-speaking subjects will be eligible since the PI and study coordinator are English-speaking.